

606028



Test Booklet No.

ENGLISH

Test Booklet Code

13

# NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

## Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below:
  - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
  - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

**Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper.** In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use Blue/Black Ball Point Pen only** for writing particulars on this page/mark responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The **CODE** for this Booklet is **13**. **Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): \_\_\_\_\_

Roll Number : In figures \_\_\_\_\_

: In words \_\_\_\_\_

Centre of Examination (in Capitals): \_\_\_\_\_

Candidate's Signature: \_\_\_\_\_ Invigilator's Signature: \_\_\_\_\_

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. The current voltage relation of diode is given by  $I = (e^{10001/T} - 1)$  mA, where the applied voltage  $V$  is in volt and the temperature  $T$  is in kelvin. If a student makes an error measuring  $\pm 0.01$  V while measuring the current of 5 mA at 300K, what will be the error in the value of current in mA?
- (1) 0.2 mA (2) 0.02 mA  
(3) 0.5 mA (4) 0.05 mA

2. Match List-I with List-II.

| List-I                                  | List-II                 |
|-----------------------------------------|-------------------------|
| (A) Young's modulus ( $Y$ )             | (I) $[ML^{-1}T^{-1}]$   |
| (B) Coefficient of viscosity ( $\eta$ ) | (II) $[ML^2T^{-1}]$     |
| (C) Planck's constant ( $h$ )           | (III) $[ML^{-1}T^{-2}]$ |
| (D) Work function ( $\phi$ )            | (IV) $[ML^2T^{-2}]$     |

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV  
(2) A-II, B-III, C-IV, D-I  
(3) A-I, B-II, C-III, D-IV  
(4) A-I, B-III, C-IV, D-II
3. Two buses  $P$  and  $Q$  start from a point at the same time and move in a straight line and their positions are represented by  $X_P(t) = \alpha t + \beta t^2$  and  $X_Q(t) = ft - t^2$ . At what time, both the buses have same velocity?
- (1)  $\frac{\alpha - f}{1 + \beta}$   
(2)  $\frac{\alpha + f}{2(\beta - 1)}$   
(3)  $\frac{\alpha + f}{2(1 - \beta)}$   
(4)  $\frac{f - \alpha}{2(1 + \beta)}$

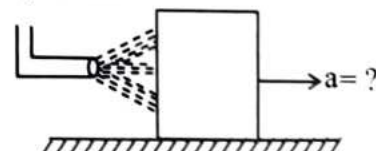
4. A particle located at  $x = 0$  at time  $t = 0$ , starts moving along the positive  $x$ -direction with a velocity  $v$  that varies as  $v = \alpha\sqrt{x}$ . The displacement of the particle varies with time as:
- (1)  $t^2$  (2)  $t$   
(3)  $t^{1/2}$  (4)  $t^3$

5. A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is:
- (1) 45 m (2) 25 m  
(3) 35 m (4) 50 m

6. A projectile is projected with velocity of 25 m/s at an angle  $\theta$  with the horizontal. After  $t$  seconds its inclination with horizontal becomes zero. If  $R$  represents horizontal range of the projectile, the value of  $\theta$  will be [use,  $g = 10 \text{ m/s}^2$ ]

- (1)  $\frac{1}{2} \sin^{-1} \left( \frac{5t^2}{4R} \right)$   
(2)  $\frac{1}{2} \sin^{-1} \left( \frac{4R}{5t^2} \right)$   
(3)  $\tan^{-1} \left( \frac{4t^2}{5R} \right)$   
(4)  $\cot^{-1} \left( \frac{R}{20t^2} \right)$

7. A block of metal weighing 2 kg is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of  $1 \text{ kg s}^{-1}$  and at a speed of  $10 \text{ ms}^{-1}$ . Then, the initial acceleration of the block, in  $\text{ms}^{-2}$ , will be:



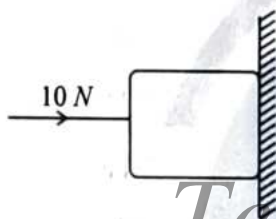
- (1) 3 (2) 6  
(3) 5 (4) 4



8. A block of mass  $M$  slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is  $\theta$ . The magnitude of the contact force will be:

- (1)  $Mg$
- (2)  $Mg \cos \theta$
- (3)  $\sqrt{Mg \sin \theta + Mg \cos \theta}$
- (4)  $Mg \sin \theta \sqrt{1 + \mu}$

9. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is:



- (1) 20 N
- (2) 50 N
- (3) 100 N
- (4) 2 N

10. The threshold frequency of a metal is  $f_0$ . When the light of frequency  $2f_0$  is incident on the metal plate, the maximum velocity of photoelectrons is  $v_1$ . When the frequency of incident radiation is increased to  $5f_0$ . The maximum velocity of photoelectrons emitted is  $v_2$ . The ratio of  $v_1$  to  $v_2$  is:

- (1)  $\frac{v_1}{v_2} = \frac{1}{4}$
- (2)  $\frac{v_1}{v_2} = \frac{1}{2}$
- (3)  $\frac{v_1}{v_2} = \frac{1}{8}$
- (4)  $\frac{v_1}{v_2} = \frac{1}{16}$

11. Mass numbers of two nuclei are in the ratio of 4 : 3. Their nuclear densities will be in the ratio of:

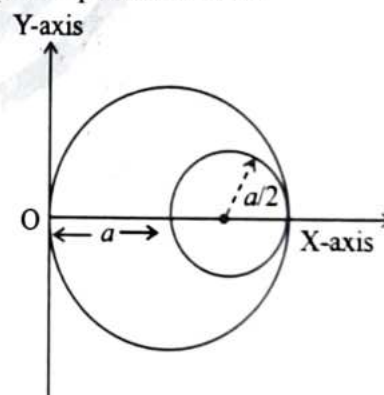
- (1) 4 : 3
- (2)  $\left(\frac{3}{4}\right)^{1/3}$
- (3) 1 : 1
- (4)  $\left(\frac{4}{3}\right)^{1/3}$

12. A particle is moving in a circular path of radius  $a$  under the action of an attractive potential  $U = -\frac{k}{2r^2}$ .

Its total energy is:

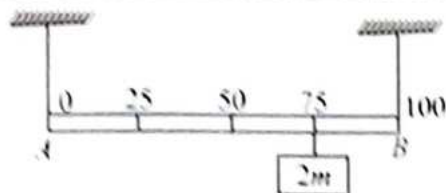
- (1)  $-\frac{k}{4a^2}$
- (2)  $-\frac{k}{2a^2}$
- (3) zero
- (4)  $\frac{3k}{2a^2}$

13. A circular hole of radius  $\left(\frac{a}{2}\right)$  is cut out of a circular disc of radius  $a$  as shown in figure. The centroid of the remaining circular portion with respect to point  $O$  will be:



- (1)  $\frac{1}{6}a$
- (2)  $\frac{10}{11}a$
- (3)  $\frac{5}{6}a$
- (4)  $\frac{2}{3}a$

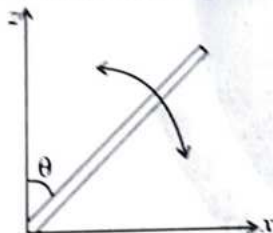
14.



Shown in the above figure, a rigid and uniform 1 m long rod  $AB$  held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass  $m$  and has another weight of mass  $2m$  hung at a distance of 75 cm from  $A$ . The tension in the string at  $A$  is:

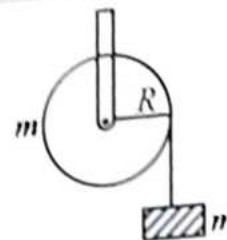
- (1)  $1\text{ mg}$
- (2)  $2\text{ mg}$
- (3)  $0.75\text{ mg}$
- (4)  $0.5\text{ mg}$

15. A slender uniform rod of mass  $M$  and length  $l$  is pivoted at one end so that it can rotate in a vertical plane (see the figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle  $\theta$  with the vertical, is



- (1)  $\frac{2g}{3l} \sin \theta$
- (2)  $\frac{3g}{2l} \cos \theta$
- (3)  $\frac{2g}{3l} \cos \theta$
- (4)  $\frac{3g}{2l} \sin \theta$

16. A mass  $m$  supported by a massless string wound around a uniform hollow cylinder of mass  $m$  and radius  $R$ . If the string does **not** slip on the cylinder, with what acceleration will the mass fall on release?



- (1)  $\frac{2g}{3}$
- (2)  $\frac{g}{2}$
- (3)  $\frac{5g}{6}$
- (4)  $g$

17. Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength  $\lambda$ . The value of principal quantum number  $n$  of the excited state will be ( $R$  = Rydberg constant)

- (1)  $\sqrt{\frac{\lambda R}{\lambda - 1}}$
- (2)  $\sqrt{\frac{\lambda R}{\lambda R - 1}}$
- (3)  $\sqrt{\frac{\lambda R}{\lambda R + 1}}$
- (4)  $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

18. Every planet revolves around the sun in an elliptical orbit.

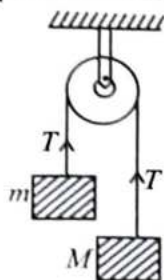
- A. The force acting on planet is inversely proportional to square of distance from sun.
- B. Force acting on planet is inversely proportional to product of the masses of the planet and the sun.
- C. The Centripetal force acting on the planet is directed away from the sun.
- D. The square of time period of revolution of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.

Choose the **most appropriate** answer from the options given below:

- (1) C and D only
- (2) A and D only
- (3) A and C only
- (4) B and C only



19. Two blocks of masses  $m$  and  $M$  are connected by means of a metal wire of cross-sectional area  $A$  passing over a frictionless fixed pulley as shown in the figure. The system is then released. If  $M = 2m$ , then the stress produced in the wire is:



- (1)  $\frac{2mg}{3A}$  (2)  $\frac{4mg}{3A}$   
 (3)  $\frac{mg}{A}$  (4)  $\frac{3mg}{4A}$
20. If 1000 droplets of water of surface tension  $0.07 \text{ N/m}$ , having same radius  $1 \text{ mm}$  each, combine to form a single drop. In the process, the released surface energy is (Take,  $\pi = \frac{22}{7}$ )
- (1)  $7.92 \times 10^{-6} \text{ J}$   
 (2)  $8.8 \times 10^{-5} \text{ J}$   
 (3)  $9.68 \times 10^{-4} \text{ J}$   
 (4)  $7.92 \times 10^{-4} \text{ J}$
21. Two different wires having lengths  $L_1$  and  $L_2$  and respective temperature coefficients of linear expansion  $\alpha_1$  and  $\alpha_2$ , are joined end-to-end. Then the effective temperature coefficient of linear expansion is:

- (1)  $\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$   
 (2)  $2\sqrt{\alpha_1 \alpha_2}$   
 (3)  $\frac{\alpha_1 + \alpha_2}{2}$   
 (4)  $4 \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2} \frac{L_2 L_1}{(L_2 + L_1)^2}$

22. When  $100 \text{ g}$  of a liquid  $A$  is at  $100^\circ\text{C}$  is added to  $50 \text{ g}$  of a liquid  $B$  at temperature  $75^\circ\text{C}$ , the temperature of the mixture becomes  $90^\circ\text{C}$ . The temperature of the mixture, if  $100 \text{ g}$  of liquid  $A$  at  $100^\circ\text{C}$  is added to  $50 \text{ g}$  of liquid  $B$  at  $50^\circ\text{C}$  will be:

- (1)  $60^\circ\text{C}$  (2)  $80^\circ\text{C}$   
 (3)  $70^\circ\text{C}$  (4)  $85^\circ\text{C}$

23. Under isothermal condition, the pressure of a gas is given by  $p = aV^{-3}$ , where  $a$  is a constant and  $V$  is the volume of the gas. The bulk modulus at constant temperature is equal to:

- (1)  $3p$  (2)  $\frac{p}{2}$   
 (3)  $2p$  (4)  $p$

24. A  $100 \Omega$  resistance, a  $0.1 \mu\text{F}$  capacitor and an inductor are connected in series across a  $250 \text{ V}$  supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is  $60 \text{ Hz}$ .

- (1)  $0.70 \text{ H}$  (2)  $70.3 \text{ mH}$   
 (3)  $7.03 \times 10^{-5} \text{ H}$  (4)  $70.3 \text{ H}$

25. Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as  $V^q$ , where  $V$  is the volume of

the gas. The value of  $q$  is  $\left( \gamma = \frac{C_p}{C_v} \right)$

- (1)  $\frac{3\gamma + 5}{6}$  (2)  $\frac{3\gamma - 5}{6}$   
 (3)  $\frac{\gamma + 1}{2}$  (4)  $\frac{\gamma - 1}{2}$

26. Curie temperature is the temperature above which:
- (1) a ferromagnetic material becomes para-magnetic.  
 (2) a paramagnetic material becomes dia-magnetic.  
 (3) a ferromagnetic material becomes dia-magnetic.  
 (4) a paramagnetic material becomes ferro-magnetic.

27. A thin convex lens made from crown glass ( $\mu = \frac{3}{2}$ ) has focal length  $f$ . When it is measured in two different liquids having refractive indices  $\frac{4}{3}$  and  $\frac{5}{3}$ . It has the focal lengths  $f_1$  and  $f_2$ , respectively. The **correct** relation between the focal lengths is:
- $f_1 = f_2 = < f$
  - $f_1 > f$  and  $f_2$  becomes negative
  - $f_2 > f$  and  $< f_1$  becomes negative
  - $f_1$  and  $f_2$  both become negative

28. When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heard. Now, some tape is attached on the prong of the fork 2. When the tuning forks are sounded again, 6 beats per second are heard.

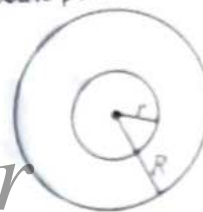
If the frequency of fork 1 is 200 Hz, then what will be the original frequency of fork 2?

- 200 Hz
  - 202 Hz
  - 196 Hz
  - 204 Hz
29. Two identical metallic spheres  $A$  and  $B$  when placed at certain distance in air repel each other with a force of  $F$ . Another identical uncharged sphere  $C$  is first placed in contact with  $A$  and then in contact with  $B$  and finally placed at mid-point between spheres  $A$  and  $B$ . The force experienced by sphere  $C$  will be:
- $3 F/2$
  - $3 F/4$
  - $F$
  - $2 F$

30. Two identical tennis balls each having mass  $m$  and charge  $q$  are suspended from a fixed point by threads of length  $l$ . What is the equilibrium separation when each thread makes a small angle  $\theta$  with the vertical?

- $d = \left( \frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{1/2}$
- $d = \left( \frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{1/3}$
- $d = \left( \frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{1/3}$
- $d = \left( \frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{1/3}$

31. A charge  $Q$  is distributed over two concentric conducting thin spherical shells radii  $r$  and  $R$  ( $R > r$ ). If the surface charge densities on the two shells are equal, the electric potential at the common centre is:



- $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{2(R^2+r^2)} Q$
- $\frac{1}{4\pi\epsilon_0} \frac{(2R+r)}{(R^2+r^2)} Q$
- $\frac{1}{4\pi\epsilon_0} \frac{(R+2r)Q}{2(R^2+r^2)}$
- $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{(R^2+r^2)} Q$

32. Two capacitors of capacities  $2C$  and  $C$  are joined in parallel and charged upto potential  $V$ . The battery is removed and the capacitor of capacity  $C$  is filled completely with a medium of dielectric constant  $K$ . The potential difference across the capacitors will now be:

- $\frac{V}{K+2}$
- $\frac{V}{K}$
- $\frac{3V}{K+2}$
- $\frac{3V}{K}$



33. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion (A):** If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the centre of earth without any variation in its magnitude.

**Reason (R):** At equator, the direction of acceleration due to the gravity is towards the centre of earth.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

34. A conducting wire of length  $l$ , area of cross-section  $A$  and electric resistivity  $\rho$  is connected between the terminals of a battery. A potential difference  $V$  is developed between its ends, causing an electric current. If the length of the wire of the same material is doubled and the area of cross-section is halved, the resultant current would be:

- (1)  $\frac{1}{4} \frac{VA}{\rho l}$
- (2)  $\frac{3}{4} \frac{VA}{\rho l}$
- (3)  $\frac{1}{4} \frac{\rho l}{VA}$
- (4)  $4 \frac{VA}{\rho l}$

35. In a meter bridge experiment, null point is obtained at 20 cm from one end of the wire when resistance  $X$  is balanced against another resistance  $Y$ . If  $X < Y$ , then where will be the new position of the null point from the same end, if one decides to balance a resistance of  $4X$  against  $Y$ ?

- (1) 50 cm
- (2) 80 cm
- (3) 40 cm
- (4) 70 cm

### Physics : Section-B (Q. No. 36 to 50)

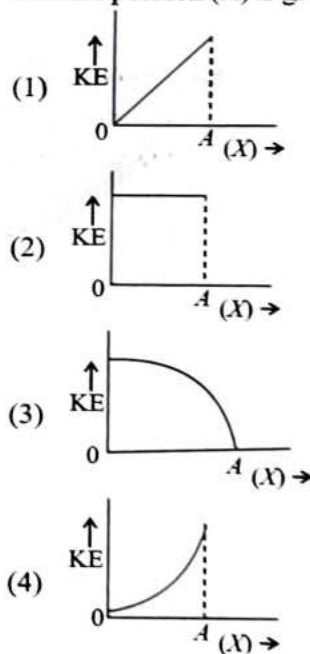
36. The electric current in a circular coil of 2 turns produces a magnetic induction  $B_1$  at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction  $B_2$  at its centre. The ratio of  $B_2/B_1$  is:

- (1)  $\frac{5}{2}$
- (2)  $\frac{25}{4}$
- (3)  $\frac{5}{4}$
- (4)  $\frac{25}{2}$

37. A proton, a deuteron and an  $\alpha$ -particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is:

- (1)  $1:\sqrt{2}:\sqrt{2}$
- (2)  $1:1:\sqrt{2}$
- (3)  $\sqrt{2}:1:1$
- (4)  $1:\sqrt{2}:1$

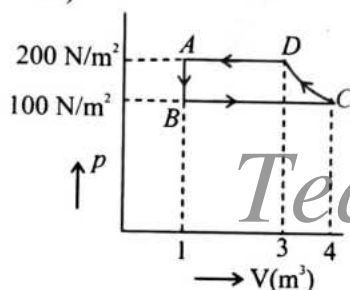
38. The variation of kinetic energy (KE) of a particle executing simple harmonic motion with the displacement ( $x$ ) starting from mean position to extreme position (A) is given by:



39. At the centre of a fixed large circular coil of radius  $R$ , a much smaller circular coil of radius  $r$  is placed. The two coils are concentric and are in the same plane. The larger coil carries a current  $I$ . The smaller coil is set to rotate with a constant angular velocity  $\omega$  about an axis along their common diameter. Calculate the emf induced in the smaller coil after a time  $t$  of its start of rotation.

(1)  $\frac{\mu_0 I}{4R} \omega \pi r^2 \sin \omega t$  (2)  $\frac{\mu_0 I}{4R} \omega r^2 \sin \omega t$   
 (3)  $\frac{\mu_0 I}{2R} \omega \pi r^2 \sin \omega t$  (4)  $\frac{\mu_0 I}{4R} \omega \pi r^2 \sin \omega t$

40. The  $p$ - $V$  diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process  $CD$  is: (use,  $\gamma = 1.4$ )



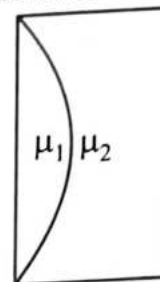
- (1) -500 J (2) -400 J  
 (3) 400 J (4) 200 J

41. In a series  $L$ - $C$ - $R$  resonance circuit, if we change the resistance only, from a lower to higher value,  
 (1) the bandwidth of resonance circuit will increase  
 (2) the resonance frequency will increase  
 (3) the quality factor will increase  
 (4) the quality factor and the resonance frequency will remain constant

42. A point source of 100 W emits light with 5% efficiency. At a distance of 5 m from the source, the intensity produced by the electric field component is:

(1)  $\frac{1}{10\pi} \frac{W}{m^2}$  (2)  $\frac{1}{20\pi} \frac{W}{m^2}$   
 (3)  $\frac{1}{2\pi} \frac{W}{m^2}$  (4)  $\frac{1}{40\pi} \frac{W}{m^2}$

43. Curved surfaces of a plano-convex lens of refractive index  $\mu_1$  and a plano-concave lens of refractive index  $\mu_2$  have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.



- (1)  $\frac{1}{\mu_2 - \mu_1}$  (2)  $\mu_1 - \mu_2$   
 (3)  $\frac{1}{\mu_1 - \mu_2}$  (4)  $\mu_2 - \mu_1$

44. The disturbance  $y(x, t)$  of a wave propagating in the positive  $x$ -direction is given by  $y = \frac{1}{1+x^2}$  at time  $t = 0$  and by  $y = \frac{1}{1+(x-1)^2}$  at  $t = 2$  s,

where  $x$  and  $y$  are in meters. The shape of the wave disturbance does **not** change during the propagation. The velocity of wave in m/s is:

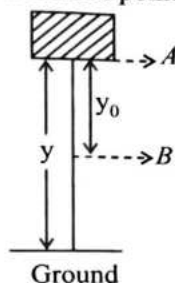
- (1) 2.0  
 (2) 4.0  
 (3) 0.5  
 (4) 1.0

45. In Young's double slit experiment, the intensity at a point where the path difference is  $\frac{\lambda}{6}$  ( $\lambda$  being the wavelength of the light used) is  $I$ . If  $I_0$  denotes the maximum intensity, then  $I/I_0$  is equal to:

- (1)  $1/\sqrt{2}$   
 (2)  $\sqrt{3}/2$   
 (3)  $1/2$   
 (4)  $3/4$



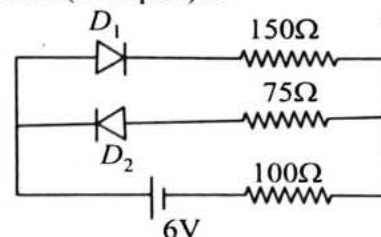
46. In the given figure, the block of mass  $m$  is dropped from the point  $A$ . The expression for kinetic energy of block when it reaches point  $B$  is:



- (1)  $\frac{1}{2}mgy_0^2$  (2)  $\frac{1}{2}mgy^2$   
 (3)  $mg(y - y_0)$  (4)  $mgy_0$
47. The de-Broglie wavelength of a proton and  $\alpha$ -particle are equal. The ratio of their velocities is:  
 (1) 4 : 3 (2) 4 : 1  
 (3) 4 : 2 (4) 1 : 4
48. Four identical solid spheres each of mass  $m$  and radius  $a$  are placed with their centres on the four corners of square of side  $b$ . The moment of inertia of the system about one side of square, where the axis of rotation is parallel to the plane of the square is:  
 (1)  $\frac{4}{5}ma^2 + 2mb^2$  (2)  $\frac{8}{5}ma^2 + mb^2$   
 (3)  $\frac{8}{5}ma^2 + 2mb^2$  (4)  $\frac{4}{5}ma^2$
49. Potential energy as a function of  $r$  is given by  $U = \frac{A}{r^{10}} - \frac{B}{r^5}$ , where  $r$  is the interatomic distance,  $A$  and  $B$  are positive constants. The equilibrium distance between the two atoms will be:

- (1)  $\left(\frac{A}{B}\right)^{1/5}$  (2)  $\left(\frac{B}{A}\right)^{1/5}$   
 (3)  $\left(\frac{2A}{B}\right)^{1/5}$  (4)  $\left(\frac{B}{2A}\right)^{1/5}$

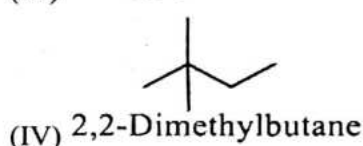
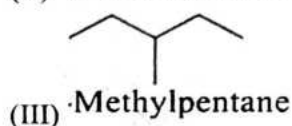
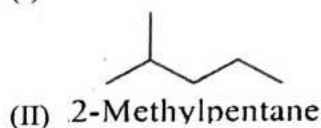
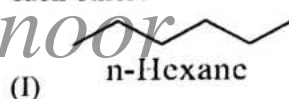
50. The circuit shown below contains two ideal diodes, each with a forward resistance of  $50\Omega$ . If the battery voltage is 6 V, the current through the  $100\Omega$  resistance (in ampere) is:



- (1) 0.027  
 (2) 0.020  
 (3) 0.030  
 (4) 0.036

## Chemistry : Section-A (Q. No. 51 to 85)

51. Which of the following are position isomer of each other?

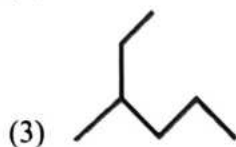
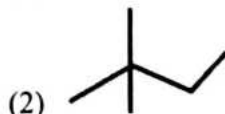
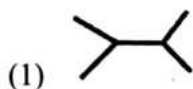


- (1) I and IV  
 (2) III and IV  
 (3) II and III  
 (4) II, III and IV
52. Which of the following is not state function?
- (1)  $q + W$   
 (2)  $q$   
 (3)  $H$   
 (4)  $H - T\Delta S$

53. Which one of the following complexes have zero crystal field stabilization energy (CFSE)?

- (1)  $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$  (2)  $[\text{Fe}(\text{CN})_6]^{4-}$   
(3)  $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$  (4)  $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

54. Which among the following compounds is optically active?



55. Which of the following statements are correct?

- A. Shape of a molecule is decided by the number of valence shell electron pairs around the central atom.  
B. Lone pair-lone pair repulsions are stronger than lone pair-bond pair repulsions.  
C. Bond pair-bond pair repulsions are stronger than lone pair-bond pair repulsions.

- (1) A, B only (2) A, C only  
(3) B, C only (4) A, B and C

56. Reaction of sodium phenoxide with  $\text{CO}_2$  in acidic medium will give;

- (1) p-cresol. (2) salicylic acid.  
(3) picric acid. (4) phenol.

57. Which of the following is a colligative property?

- (1) Boiling point  
(2) Vapour pressure  
(3) Osmotic pressure  
(4) Freezing point

58. Which of the following amines is the most basic?

- (1) Aniline  
(2) Ethylamine  
(3) p-Nitroaniline  
(4) Diphenylamine

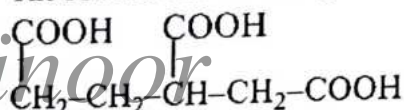
59. What is the mass of sodium carbonate required to prepare 400 mL of 0.275 M aqueous solution?

- (1) 5.83 g (2) 11.66 g  
(3) 17.49 g (4) 18.20 g

60. For a first order reaction;  $\text{A} \rightarrow \text{B}$ , the rate of reaction at the reactant concentration of 0.02M is found to be  $6.0 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$ . The half-life period of the reaction is;

- (1) 2.31 sec (2) 34.3 sec  
(3) 69.3 sec (4) 23.1 sec

61. The IUPAC name of the given compound is;



- (1) 2-Carboxyhexane-1,6-dioic acid  
(2) Butane-1,3,4-tricarboxylic acid  
(3) Butane-1,2,4-tricarboxylic acid  
(4) 4-Carboxyhexane-1,6-dioic acid

62. Given below are two statements:

**Statement I:** The mass of one mole of substance in grams is called its molar mass.

**Statement II:** The temperature of two scales are related to each other by the relation:

$$^{\circ}\text{F} = \frac{9}{5}(^{\circ}\text{C}) + 32$$

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.  
(2) Statement I is incorrect but Statement II is correct.  
(3) Both Statement I and Statement II are correct.  
(4) Both Statement I and Statement II are incorrect.



63. The correct order of the atomic radii is;

- (1)  $\text{Na}^+ < \text{Al}^{+3} < \text{Mg}^{+2}$
- (2)  $\text{Na}^+ < \text{Mg}^{+2} < \text{Al}^{+3}$
- (3)  $\text{Cl}^- < \text{P}^{3-} < \text{S}^{2-}$
- (4)  $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$

64. The specific conductivity of 1N KCl solution is  $0.029 \Omega^{-1}\text{cm}^{-1}$ . If the cell constant of the cell is  $0.1 \text{ cm}^{-1}$  then conductance will be;

- (1)  $1.10 \Omega^{-1}$  (2)  $0.29 \Omega^{-1}$
- (3)  $0.56 \Omega^{-1}$  (4)  $2.80 \Omega^{-1}$

65. The Cannizzaro reaction is not given by;

- (1) trimethyl acetaldehyde
- (2) formaldehyde
- (3) benzaldehyde
- (4) acetaldehyde

66. Match List-I with List-II.

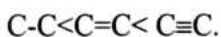
| List-I<br>(Quantum Number)   | List-II<br>(Value)                 |
|------------------------------|------------------------------------|
| (A) Azimuthal quantum number | (I) $n=1, 2, 3, \dots$             |
| (B) Magnetic quantum number  | (II) 0 to $(n-1)$                  |
| (C) Spin quantum number      | (III) $-\frac{1}{2}, +\frac{1}{2}$ |
| (D) Principal quantum number | (IV) $-l$ to $+l$                  |

Choose the correct answer from the options given below.

- (1) A-II, B-IV, C-III, D-I
- (2) A-IV, B-II, C-III, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-III, C-II, D-I

67. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Bond energy order:



**Reason R:** Bond energy increases with increase in bond order.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

68. Match List-I with List-II.

| List-I<br>(Species)                  | List-II<br>(Oxidation state of central atom) |
|--------------------------------------|----------------------------------------------|
| (A) $\text{IF}_5$                    | (I) +6                                       |
| (B) $\text{CO}_2$                    | (II) -2                                      |
| (C) $\text{S}^{2-}$                  | (III) +4                                     |
| (D) $\text{H}_2\text{S}_2\text{O}_8$ | (IV) +5                                      |

Choose the correct answer from the options given below.

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

69. The change in entropy when 25.0 g of water condenses from a gas to a liquid at the normal boiling point of water ( $100^\circ\text{C}$ ) is nearly.

$$\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$$

- (1) +161 J/K
- (2) -151 J/K
- (3) +145 J/K
- (4) -245 J/K

70. A gas has a vapour density 11.2. The volume occupied by 1 g of that gas at STP is:

- (1) 1L
- (2) 11.2 L
- (3) 22.4 L
- (4) 4 L

71. Given below are two statements:

**Statement I:** Bohr's theory explains the spectrum of He atom.

**Statement II:** Bohr's theory was unable to explain the splitting of spectral lines in the presence of magnetic field and electric field.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

72. Which of the following are essential amino acids?

- |                   |                   |
|-------------------|-------------------|
| A. Arginine,      | B. Phenylalanine, |
| C. Aspartic acid, | D. Cysteine,      |
| E. Histidine,     | F. Valine,        |
| G. Proline        |                   |

- (1) A, B, C and D
- (2) B, C, D and E
- (3) C, D, E and F
- (4) A, B, E and F

73. Which of the following compound **cannot** give a test for nitrogen by Lassaigne's extract?

- |             |                   |
|-------------|-------------------|
| (1) Aniline | (2) Hydroxylamine |
| (3) Glycine | (4) Urea          |

74. Which of the following statement (s) are correct?

- A. In group 15 elements, covalent and ionic radii increases down the group but there is small increase from As to Bi.
- B. Ionisation enthalpy of group 15 element is less than that of 14 elements, in corresponding period.
- C. In group 15 elements, electronegativity decreases down the group with increasing atomic size.
- D. In group 15 elements, metallic character increases down the group.

- |                |                  |
|----------------|------------------|
| (1) A, B and C | (2) Only A and D |
| (3) A, C and D | (4) B, C and D   |

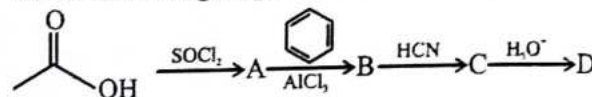
75. Match List-I with List-II.

| List-I<br>(Types of solution) | List-II<br>(Example)                    |
|-------------------------------|-----------------------------------------|
| (A) Gas in liquid             | (I) Hydrogen gas in palladium           |
| (B) Solid in liquid           | (II) Oxygen dissolved in water          |
| (C) Liquid in gas             | (III) Glucose dissolved in water        |
| (D) Gas in solid              | (IV) Chloroform mixed with nitrogen gas |

Choose the **correct** answer from the options given below.

- |                            |
|----------------------------|
| (1) A-II, B-III, C-I, D-IV |
| (2) A-II, B-III, C-IV, D-I |
| (3) A-III, B-II, C-I, D-IV |
| (4) A-III, B-II, C-IV, D-I |

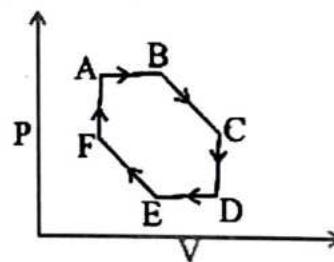
76. In the following sequence of reactions,



The **correct** IUPAC name of product D is;

- (1) 2-Hydroxy-2-phenylpropanoic acid
- (2) 2-Hydroxy-1-phenylpropanoic acid
- (3) 1-Hydroxy-2-phenylpropanoic acid
- (4) 2-Hydroxy-2-methylbenzoic acid

77. Different steps are involved in the following cyclic process. The step representing isobaric process is;



- |           |           |
|-----------|-----------|
| (1) A → B | (2) B → C |
| (3) E → F | (4) F → A |



78. Given below are two statements:

**Statement I:** Colour of transition metal complexes can be explained by d-d transitions.

**Statement II:**  $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$  and  $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$  show hydrate isomerism.

In the light of the above statements, choose the **correct** answer from the options given below:

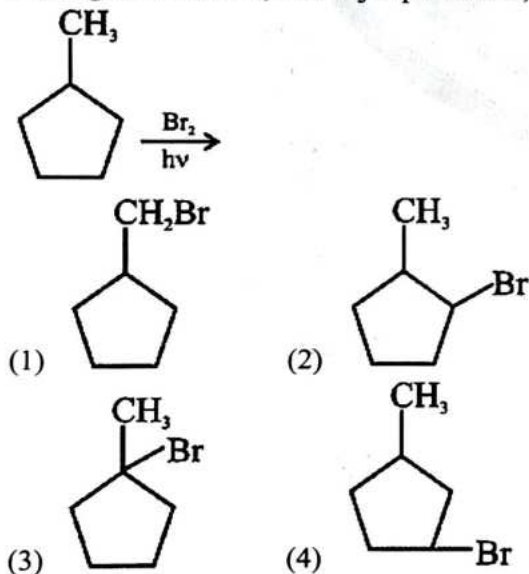
- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

79. Half life time for a first order reaction is 15 min.

The time required for 99.9% completion of the reaction is;

- (1) 4.5 hr
- (2) 1.5 hr
- (3) 2.8 hr
- (4) 2.5 hr

80. In the given reaction, the major product is;



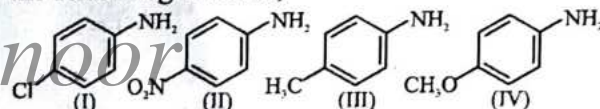
81. Match List-I with List-II.

| List-I<br>(Species)   | List-II<br>(Shape)  |
|-----------------------|---------------------|
| (A) $\text{ICl}_2^-$  | (I) Linear          |
| (B) $\text{BrF}_2^+$  | (II) Tetrahedral    |
| (C) $\text{ClF}_4^-$  | (III) Square planar |
| (D) $\text{AlCl}_4^-$ | (IV) Angular        |

Choose the **correct** answer from the options given below.

- (1) A-I, B-II, C-IV, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-I, B-IV, C-III, D-II
- (4) A-III, B-I, C-II, D-IV

82. The **correct** order of increasing basic nature of the following bases is;



- (1)  $\text{II} < \text{IV} < \text{III} < \text{I}$
- (2)  $\text{II} < \text{I} < \text{III} < \text{IV}$
- (3)  $\text{I} < \text{III} < \text{IV} < \text{II}$
- (4)  $\text{II} < \text{I} < \text{IV} < \text{III}$

83. Given below are two statements.

**Statement I:** Bond angles of  $\text{BF}_3$  and  $\text{NF}_3$  are different.

**Statement II:** The dipole moment of  $\text{BF}_3$  is zero.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

84. Given below are two statements.

**Statement I:** The spin only magnetic moment of  $\text{Sc}^{3+}$  is 1.73 BM.

**Statement II:** The spin only magnetic moment (in BM) of an ion is equal to  $\sqrt{n(n+2)}$ , where,  $n$  is the number of unpaired electrons present in the ion.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

85. Vitamin that can not be stored in our body is;

- (1) Vitamin D
- (2) Vitamin B<sub>6</sub>
- (3) Vitamin B<sub>12</sub>
- (4) Vitamin K

**Chemistry : Section-B (Q. No. 86 to 100)**

86. A basic buffer solution contains equal concentration of base and its salt. If the value of  $K_b$  is  $10^{-5}$ , then the pH of the buffer solution would be;

- (1) 11
- (2) 10
- (3) 12
- (4) 9

87. Zr ( $Z = 40$ ) and Hf ( $Z = 72$ ) have similar atomic and ionic radii because of;

- (1) Diagonal relationship
- (2) Lanthanoid contraction
- (3) Having similar chemical properties
- (4) Belonging to the same group

88. Given below are two statements.

**Statement I:** The relative ease of dehydration of alcohols follow the following order:

Tertiary > Secondary > Primary

**Statement II:** Formation of carbocation is the slowest step of the reaction.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

89. Which of the following materials are semiconductors?

- A. Germanium
- B. Graphite
- C. Silicon
- D. Copper oxide

- (1) (A), (B) and (D)
- (2) (B), (C) and (D)
- (3) (A), (C) and (D)
- (4) (B), (C) and (D)

90. The time required for the completion of 80% of the reaction, if the rate constant for a reaction is ' $\alpha$ '; ( $\log 5 = 0.7$ )

- (1)  $1.60/\alpha$
- (2)  $2.30/\alpha$
- (3)  $4.60/\alpha$
- (4)  $0.693/\alpha$

91. For a reversible reaction:  $\text{A(s)} + 2\text{B(g)} \rightleftharpoons 3\text{C(g)}$  at any temperature 'T' relation between  $K_p$  and  $K_c$  is;

$K_c = K_p(RT)^x$ , hence value of ' $x$ ' will be;

- (1) -1
- (2) 1
- (3) 0
- (4) 2

92. Incorrect statement regarding interstitial compounds?

- (1) They have high melting points than those of pure metals.
- (2) They are very hard.
- (3) They retain metallic conductivity.
- (4) They are chemically reactive.



93. Which of the following is the **correct** decreasing order of dipole moment?

- (1)  $\text{CH}_3\text{F} > \text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
- (2)  $\text{CH}_3\text{F} > \text{CH}_3\text{Cl} > \text{CH}_3\text{I} > \text{CH}_3\text{Br}$
- (3)  $\text{CH}_3\text{Cl} > \text{CH}_3\text{F} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
- (4)  $\text{CH}_3\text{F} < \text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{I}$

94. Which of the following is **correct** order of energy of atomic orbital for H-atom;

- (1)  $3d > 4s$
- (2)  $3s < 3p$
- (3)  $2s < 2p$
- (4)  $4s = 4p$

95. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

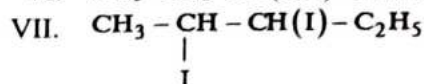
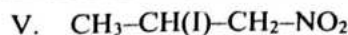
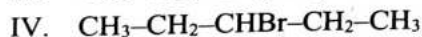
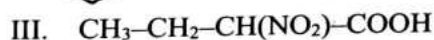
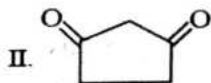
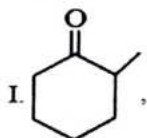
**Assertion A:** Functional isomers have different functional groups but the same molecular formula.

**Reason R:** They exhibit similar chemical and physical properties due to identical molecular formulas.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

96. Which of the following compounds contain achiral carbon atoms.



- (1) Both I & III
- (2) Both II and IV
- (3) Both III and V
- (4) Both VI and VII

97. Which of the following reactions does not produce an amine?

- (1) Hoffmann bromamide degradation
- (2) Gabriel phthalimide synthesis
- (3) Reduction of nitro compounds
- (4) Friedel-Crafts acylation

98. In Duma's method for estimation of nitrogen 0.25g of an organic compound gave 40ml of nitrogen collected at 300K temperature and 730 mm pressure. If the aqueous tension at 300K is 30 mm, the percentage of nitrogen in the compound is;

- (1) 16.76%
- (2) 31.25%
- (3) 25.25%
- (4) 20.25%

99. Which of the following is the correct decreasing order of electronegativity?

- (1)  $\text{B} > \text{Al} > \text{Tl} > \text{Ga}$
- (2)  $\text{Al} > \text{B} > \text{Tl} > \text{Ga}$
- (3)  $\text{B} > \text{Al} > \text{Ga} > \text{Tl}$
- (4)  $\text{B} > \text{Tl} > \text{Ga} > \text{Al}$

100. Consider the reaction,  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ ;  $\Delta H = -93.6 \text{ kJ mol}^{-1}$  the number of moles of  $\text{N}_2$  at equilibrium will increase when;

- (1) pressure is decreased
- (2) temperature is decreased
- (3) argon gas is added at constant volume
- (4)  $\text{NH}_3$  is removed

**Botany : Section-A (Q. No. 101 to 135)**

101. 'Red tides' are caused by;  
 (1) dinoflagellates. (2) euglenoids.  
 (3) chrysophytes. (4) protozoans.
102. Identify the **mismatched** pair.  
 (1) Diptera - Order  
 (2) Dicotyledonae - Class  
 (3) Sapindales - Family  
 (4) Angiospermae - Division
103. Given below are two statements:  
**Statement I:** Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence.  
**Statement II:** The stems are unbranched in *Cycas* and branched in *Cedrus*.  
 In the light of the above statements, choose the most appropriate answer from the options given below:  
 (1) Statement I is correct but Statement II is incorrect.  
 (2) Statement I is incorrect but Statement II is correct.  
 (3) Both Statement I and Statement II are correct.  
 (4) Both Statement I and Statement II are incorrect.
104. All tissues on the inner side of the \_\_\_\_\_ constitute the stele.  
 (1) endodermis  
 (2) pericycle  
 (3) vascular bundles  
 (4) pith
105. If '+' represents beneficial interaction, '-' represents detrimental interaction and '0' represents neutral interaction, then the population interaction amensalism will be indicated as;  
 (1) +, - (2) +, +  
 (3) -, 0 (4) +, 0

106. The first action spectrum of photosynthesis was described by:  
 (1) Cornelius van Niel.  
 (2) Jan Ingenhousz.  
 (3) Julius von Sachs.  
 (4) T.W. Engelmann.

107. Match List-I and List-II.

| List-I<br>(Cell organelles)     | List-II<br>(Characteristic)                                |
|---------------------------------|------------------------------------------------------------|
| (A) Rough endoplasmic reticulum | (I) Cisternae are concentrically arranged near the nucleus |
| (B) Golgi apparatus             | (II) Inner membrane forms a cristae                        |
| (C) Lysosomes                   | (III) Continuous with the outer membrane of the nucleus    |
| (D) Mitochondria                | (IV) Membrane bound vesicular structures                   |

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV  
 (2) A-III, B-IV, C-I, D-II  
 (3) A-IV, B-I, C-III, D-II  
 (4) A-III, B-I, C-IV, D-II

108. Identify the **mismatched** pair.  
 (1) F.W. Went - Isolation of auxin.  
 (2) E. Kurosawa - Worked on fungal disease of rice seedlings.  
 (3) F. Skoog - Crystallised cytokinesis promoting active substances.  
 (4) H.H. Cousins - Confirmed the release of a volatile substance from ripened oranges.





109. \_\_\_\_\_ noted that the behaviour of chromosomes was parallel to the behaviour of genes and used chromosome movement to explain Mendel's laws.

- (1) T.H. Morgan
- (2) Watson and Crick
- (3) Bateson and Punnett
- (4) Sutton and Boveri

110. Which of the following components are essential for the tissue culture medium?

- (a) Sucrose
- (b) Inorganic salts
- (c) Antibiotics
- (d) Vitamins and amino acids
- (e) Cytokinins and auxins

Choose the correct answer from the options given below:

- (1) (a), (b), (c) and (d)
- (2) (a), (b) and (e)
- (3) (a), (b), (d) and (e)
- (4) (a), (c) and (e)

111. In the Citric acid cycle, during the conversion of \_\_\_\_\_ a molecule of GTP is synthesised.

- (1) succinyl-CoA to succinic acid
- (2) citric acid to  $\alpha$ -ketoglutaric acid
- (3) succinic acid to malic acid
- (4) malic acid to oxaloacetic acid

112. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Histones are organised to form a unit of eight molecules called histone octamer.

**Reason R:** The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

113. Match List-I and List-II.

**List-I**

**List-II**

- |                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| (A) Complex I   | (I) Transfer of electrons to cytochrome c via cytochrome bc <sub>1</sub> complex  |
| (B) Complex II  | (II) Complex containing cytochromes a and a <sub>3</sub> , and two copper centers |
| (C) Complex III | (III) Electrons from NADH are oxidized                                            |
| (D) Complex IV  | (IV) Oxidation of succinate to fumarate                                           |

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-IV, D-II

114. Identify the **incorrect** statement.

- (1) In dicotyledonous stems, cambium is present between phloem and xylem.
- (2) Monocotyledonous stems form secondary tissues.
- (3) In leaves, xylem and phloem are situated at the same radius.
- (4) In stem, phloem is usually located only on the outer side of xylem.

115. \_\_\_\_\_ of an ecosystem is the rate of production of organic matter during photosynthesis.

- (1) Net primary productivity
- (2) Gross primary productivity
- (3) Secondary productivity
- (4) Primary productivity

116. Given below are two statements:

**Statement I:** In stroma of the chloroplast, enzymatic reactions synthesise sugar, which in turn forms starch.

**Statement II:** There is a clear division of labor within the chloroplast.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

117. In eukaryotes, RNA polymerase II transcribes;

- (1) hnRNA.
- (2) tRNA.
- (3) snRNA.
- (4) rRNA.

118. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** The zone, proximal to region of elongation in root, is called region of maturation.

**Reason R:** The cells of the elongation zone of root gradually differentiate and mature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

119. Choose the **incorrect** statement w.r.t the stages of mitosis.

- (1) During prophase, each centrosome radiates out microtubules called asters.
- (2) The chromosomes are spread through the cytoplasm of the cell during metaphase.
- (3) During anaphase, chromosomes move to the equator.
- (4) Final stage of karyokinesis is the telophase.

120. All of the following are *in situ* conservation strategies, **except**;

- (1) biosphere reserves.
- (2) National parks.
- (3) sacred groves.
- (4) wildlife safari parks.

121. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Carboxylation is the fixation of  $\text{CO}_2$  into a stable organic intermediate.

**Reason R:** Carboxylation is the most crucial step of the Calvin cycle where  $\text{O}_2$  is utilised for the carboxylation of RuBP.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

122. Match the columns (characters with dominant contrasting trait) and find out the correct combination:

| Column-I                                   | Column-II      |
|--------------------------------------------|----------------|
| (a) Pod shape                              | (i) Round      |
| (b) Pod color                              | (ii) Green     |
| (c) Seed shape                             | (iii) Inflated |
| (d) Seed colour                            | (iv) Yellow    |
| (1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) |                |
| (2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv) |                |
| (3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv) |                |
| (4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii) |                |



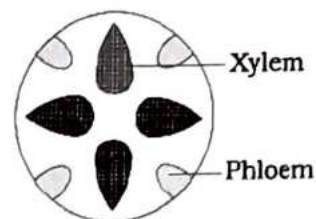


123. Choose the **incorrect** statement w.r.t the adapter molecule.
- (1) Francis Crick postulated the presence of an adapter molecule.
  - (2) There are no tRNAs for stop codons.
  - (3) tRNA has an amino acid acceptor end to which it binds to amino acids.
  - (4) tRNAs are specific for each DNA strand.
124. Choose the **incorrect** statement w.r.t the cyclic photophosphorylation.
- (1) Only PS I is functional.
  - (2) A possible location is the stroma lamellae.
  - (3) Results in the synthesis of ATP and NADPH +  $H^+$ .
  - (4) Occurs when only light of wavelengths beyond 680 nm are available for excitation.
125. The basal body of cilia is made up of;
- (1) centromere.
  - (2) centrioles.
  - (3) kinetochore.
  - (4) axoneme.
126. Identify the **mismatched** pair.
- (1) Dicot root - Presence of casparian strips
  - (2) Monocot root - Polyarch xylem bundles
  - (3) Dicot stem - Presence of starch sheath
  - (4) Monocot stem - Phloem parenchyma is present
127. Arrange the following steps in the correct order as they occur in glycolysis:
- (a) Conversion of glucose to glucose-6-phosphate
  - (b) Conversion of phosphoenolpyruvate to pyruvic acid
  - (c) Splitting of fructose-1,6-bisphosphate into triose phosphates
  - (d) Oxidation of glyceraldehyde-3-phosphate to 1, 3-bisphosphoglyceric acid
  - (e) 3-phosphoglycerate to 2-phosphoglycerate

Choose the correct sequence from the options given below:

- (1) (a)-(c)-(d)-(e)-(b)
- (2) (a)-(b)-(c)-(d)-(e)
- (3) (c)-(a)-(b)-(d)-(e)
- (4) (d)-(a)-(b)-(c)-(e)

128. The type of vascular bundle shown below is generally found in;



- (1) stems.
- (2) leaves.
- (3) roots.
- (4) stems and leaves.

129. Identify the example of species diversity of living organisms.

- (1) *Rauwolfia vomitoria* growing in different Himalayan ranges.
- (2) Varieties of mango in India.
- (3) Amphibian diversity is greater in the Western Ghats than in the Eastern Ghats.
- (4) India's deserts exhibit more diverse natural habitats compared to Scandinavian countries like Norway.

130. Identify the **incorrect** statement.

- (1) DNA ligase joins the fragments of new fragments of DNA formed on the template DNA with polarity  $3' \rightarrow 5'$ .
- (2) DNA-dependent DNA polymerase uses DNA as template for replication.
- (3) On the template DNA strand with polarity  $3' \rightarrow 5'$ , the replication is continuous.
- (4) Origin of replication is a specific region of DNA where replication originates.

131. Which of the following are examples of *ex-situ* conservation methods?

- (a) Cryopreservation
- (b) National parks
- (c) Biosphere reserve
- (d) Tissue culture
- (e) Botanical gardens

- (1) (a), (b), (c)      (2) (b), (c), (e)  
(3) (a), (d), (e)      (4) (c), (d), (e)

132. Match the following with respect to meiosis.

| Column-I       | Column-II           |
|----------------|---------------------|
| (A) Zygotene   | (i) Terminalization |
| (B) Pachytene  | (ii) Chiasmata      |
| (C) Diplotene  | (iii) Crossing over |
| (D) Diakinesis | (iv) Synapsis       |

Select the correct option from the following:

- (1) A-(iv), B-(iii), C-(ii), D-(i)
- (2) A-(i), B-(ii), C-(iv), D-(iii)
- (3) A-(ii), B-(iv), C-(iii), D-(i)
- (4) A-(iii), B-(iv), C-(i), D-(ii)

133. Identify the **incorrect** statement.

- (1) The beginning of diplotene is recognised by the dissolution of the synaptonemal complex.
- (2) Recombined homologous chromosomes of the bivalents separate from each other except at the sites of crossovers during diplotene stage.
- (3) Site of crossover forms a structure called chiasmata.
- (4) In oocytes of some invertebrates, diplotene can last for seconds.

134. Given below are two statements:

**Statement I:** Mendel crossed the tall plant from  $F_2$  with a dwarf plant to determine the genotype of the short plants.

**Statement II:** In a typical test cross, an organism showing a dominant phenotype (and whose genotype is to be determined) is crossed with the recessive parent instead of self-crossing.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

135. Identify the nitrogenous base which is **not** required for the synthesis of DNA.

- (1) Cytosine      (2) Uracil
- (3) Thymine      (4) Adenine

**Botany : Section-B (Q. No. 136 to 150)**

136. Streptokinase is a bioactive molecule produced by;

- (1) *Trichoderma*.
- (2) *Streptococcus*.
- (3) *Clostridium*.
- (4) *Monascus*.

137. Read the following statements.

- A. In symmetry, the flower may be actinomorphic (bilateral symmetry) or zygomorphic (radial symmetry).
- B. *Cassia* shows zygomorphic symmetry.
- C. A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre.
- D. Mustard shows actinomorphic symmetry.

Choose the *most appropriate* answer from the options given below:

- (1) A and B are incorrect
- (2) B, C and D are correct
- (3) A and C are correct
- (4) A, B and D are incorrect



138. Identify the **correct** statement w.r.t reproduction in algae.

- (1) Fusion of two gametes dissimilar in size is observed in case of *Spirogyra*.
- (2) Fusion of flagellated gametes similar in size takes place in *Eudorina*.
- (3) Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete is termed anisogamy.
- (4) Fusion of non-flagellated gametes similar in size is called isogamy.

139. R. H. Whittaker used all the following criteria for classification of organisms, **except**;

- (1) cell structure.
- (2) mode of nutrition.
- (3) presence of cell membrane.
- (4) phylogenetic relationships.

140. Match List-I and List-II.

| List-I |         | List-II |                               |
|--------|---------|---------|-------------------------------|
| (A)    | Exon    | (I)     | Non-coding sequence           |
| (B)    | Intron  | (II)    | Coding sequence               |
| (C)    | Capping | (III)   | Polyadenylation               |
| (D)    | Tailing | (IV)    | Methyl guanosine triphosphate |

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-IV, C-I, D-III

141. Arrange the following trophic levels in the correct sequence from lowest to highest in dry weights in a terrestrial ecosystem

- (a) Tertiary Consumers
- (b) Primary Consumers
- (c) Secondary Consumers
- (d) Producers

Options:

- (1) (d), (c), (b), (a)
- (2) (a), (c), (b), (d)
- (3) (c), (b), (a), (d)
- (4) (b), (a), (d), (c)

142. Read the following statements carefully and select the **correct** ones.

- A. Inclusion bodies are not bound by any membrane system and lie free in the nucleoplasm.
- B. All eukaryotic cells are not identical.
- C. Both the cilium and flagellum emerge from centrioles called the basal bodies.
- D. The materials to be packaged from ER fuse with the trans face of the Golgi apparatus and move towards the maturing face.

- (1) A and C
- (2) C and D
- (3) B and C
- (4) A and D

143. Match List-I and List-II.

| List-I |             | List-II |                                                      |
|--------|-------------|---------|------------------------------------------------------|
| (A)    | Scutellum   | (I)     | Radical enclosed in an undifferentiated sheath       |
| (B)    | Coleoptile  | (II)    | Terminates with the stem tip                         |
| (C)    | Coleorrhiza | (III)   | Large shield-shaped structure                        |
| (D)    | Epicotyl    | (IV)    | Leaf primordia enclosed in a hollow foliar structure |

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

144. Match the columns and choose the correct combination

| Column-I |             | Column-II |                               |
|----------|-------------|-----------|-------------------------------|
| (a)      | Funicle     | (i)       | Small opening of ovule        |
| (b)      | Integuments | (ii)      | Stalk of ovule                |
| (c)      | Chalaza     | (iii)     | Protective envelopes of ovule |

- (d) Hilum (iv) Junction part of ovule and stalk  
(e) Micropyle (v) Basal part of ovule

- (1) (a)-(ii), (b)-(iii), (c)-(v), (d)-(iv), (e)-(i)  
(2) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv), (e)-(v)  
(3) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv), (e)-(v)  
(4) (a)-(ii), (b)-(iv), (c)-(v), (d)-(i), (e)-(iii)

145. The green expanded part of the leaf with veins and veinlets is;

- (1) pulvinus. (2) leaf base.  
(3) petiole. (4) leaf blade.

146. Identify the **incorrect** statement w.r.t  $G_0$  phase.

- (1) It is also known as quiescent stage.  
(2) Cell of this stage remain metabolically inactive.  
(3) There is no longer proliferation in the cells which are present in this stage, unless called to do so.  
(4) Heart cells in animals exhibit this stage.

147. Match the columns and find out the correct combination:

| Column-I                                                 |       | Column-II                                  |  |
|----------------------------------------------------------|-------|--------------------------------------------|--|
| (a) Invertase                                            | (i)   | Pyruvate $\rightarrow$ Lactate             |  |
| (b) Hexokinase                                           | (ii)  | Pyruvate $\rightarrow$ Ethanol + $CO_2$    |  |
| (c) Pyruvic acid decarboxylase and alcohol dehydrogenase | (iii) | Sucrose $\rightarrow$ Glucose + Fructose   |  |
| (d) Lactate dehydrogenase                                | (iv)  | Glucose $\rightarrow$ Glucose-6-P          |  |
| (e) Citrate synthase                                     | (v)   | Acetyl CoA + OAA $\rightarrow$ Citric acid |  |
| (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii), (e)-(v)      |       |                                            |  |
| (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i), (e)-(v)      |       |                                            |  |
| (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i), (e)-(v)      |       |                                            |  |
| (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(v), (e)-(iv)      |       |                                            |  |

148. Read the following statements.

- A. A population at any given time is composed of individuals of different ages.  
B. The size of a population for any species is a static parameter.  
C. Immigration is the number of individuals of the population who left the habitat and gone elsewhere during a particular time period.

Choose the *most appropriate* answer from the options given below:

- (1) A and B are correct and C is incorrect.  
(2) A and C are correct and B is incorrect.  
(3) A is correct and B and C are incorrect.  
(4) B is correct and A and C are incorrect.

149. Which of the following shows the floral characteristic of Cruciferae?

- (1) Six tepals, zygomorphic, six stamens, bilocular ovary, axile placentation.  
(2) Hermaphrodite, actinomorphic, tetradynamous, hypogynous, bicarpellary, bilocular (replum), placentation parietal.  
(3) Bisexual, actinomorphic, six stamens, polyandrous, superior ovary, axile placentation.  
(4) Three tepals, bisexual, zygomorphic, gamophyllous, inferior ovary, marginal placentation.

150. Match the columns and find out the correct combination:

| Column-I                                                    |       | Column-II  |  |
|-------------------------------------------------------------|-------|------------|--|
| (a) Independent pairs segregate independently of each other | (i)   | Chromosome |  |
| (b) Association of two genes                                | (ii)  | Factors    |  |
| (c) One pair segregates independently of another pair       | (iii) | Linkage    |  |
| (d) Characters controlled by discrete units                 | (iv)  | Gene       |  |



- (1) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
- (3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- (4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

### Zoology : Section-A (Q. No. 151 to 185)

151. What will be the  $pO_2$  and  $pCO_2$  in atmospheric air as compared to alveoli?

- (1) High  $pO_2$  and low  $pCO_2$
- (2) High  $pCO_2$  and low  $pO_2$
- (3) Both are equal ( $pCO_2 = pO_2$ )
- (4) Low  $pO_2$  and low  $pCO_2$

152. In 1953, S. L. Miller created the primitive earth conditions in the laboratory and gave experimental evidence for the origin of first form of life from pre-existing non-living organic molecules. The primitive earth conditions created include:

- (1) low temperature, volcanic storms, atmosphere rich in oxygen.
- (2) low temperature, volcanic storms, reducing atmosphere.
- (3) high temperature, volcanic storms, non-reducing atmosphere.
- (4) high temperature, volcanic storms, reducing atmosphere containing  $CH_4$ ,  $NH_3$ , etc.

153. Match List-I with List-II:

- | List-I            | List-II                      |
|-------------------|------------------------------|
| (A) PCR           | (I) Restriction endonuclease |
| (B) <i>EcoRI</i>  | (II) Vector                  |
| (C) pBR322        | (III) Bacterium              |
| (D) <i>E.coli</i> | (IV) Gene amplification      |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-III, C-II, D-IV
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-II, C-I, D-IV

154. Expiration occurs due to:

- (1) relaxation of diaphragm and intercostal muscles.
- (2) contraction of internal intercostal muscles and diaphragm.
- (3) relaxation of abdominal muscles only.
- (4) contraction of diaphragm and relaxation of abdominal muscles.

155. Restriction enzymes are used in genetic engineering because:

- (1) they can join different DNA fragments.
- (2) they can cleave DNA at a specific target.
- (3) they are nucleases that cut DNA at variable sites.
- (4) they are proteolytic enzymes which can degrade harmful enzymes.

156. Select the **correct** match.

- (1) Polymerase chain reaction  $\rightarrow$  Antigen- antibody interaction
- (2) GEAC  $\rightarrow$  Genetic engineering approval committee
- (3) ELISA  $\rightarrow$  Enzyme lysed immuno- sorbent assay
- (4) GMO  $\rightarrow$  Genetically modified organs

157. Given below are two statements:

**Statement I:** Primary metabolites have identifiable functions and play known roles in normal physiological processes.

**Statement II:** Secondary metabolites have ecological importance.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

158. Milk secreted from the cells of alveoli of mammary lobes reaches nipple. Which of the following is the **correct** sequence of transport of milk?

- (1) Mammary tubule → Mammary duct → Mammary ampulla → Lactiferous duct
- (2) Mammary duct → Mammary tubule → Lactiferous duct → Mammary ampulla
- (3) Mammary duct → Mammary tubule → Mammary ampulla → Lactiferous duct
- (4) Mammary ampulla → Mammary tubule → Mammary duct → Lactiferous duct

159. Select the **correct** match of animals with their common names.

- (1) *Euspongia* - Fresh water sponge
- (2) *Meandrina* - Sea anemone
- (3) *Fasciola* - Tapeworm
- (4) *Ancylostoma* - Hookworm

160. Given below are two statements:

**Statement I:** In unicellular organisms, all functions are performed by a single cell.

**Statement II:** In complex multicellular animals, the distinct functions are carried out by different groups of cells.

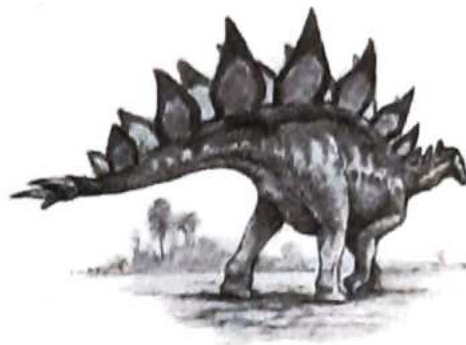
In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

161. Which of the following is preferably used to deliver a desirable gene into plant cells?

- (1) Retrovirus
- (2) *Agrobacterium tumefaciens*
- (3) Plasmid pBR322
- (4) None of these

162. Identify the given diagram.



- (1) *Therapsida*
- (2) *Brachiosaurus*
- (3) *Tyrannosaurus*
- (4) *Stegosaurus*

163. What is the amount of haemoglobin present in 100 ml of blood?

- |               |               |
|---------------|---------------|
| (1) 20-24 gms | (2) 40-44 gms |
| (3) 12-16 gms | (4) 30-34 gms |

164. Read the following statements.

- A. Hormones are also secreted by some tissues which are not endocrine glands.
- B. Hormones are non-nutrient chemicals.
- C. Hormones act as intracellular messengers and are produced in trace amounts.

Choose the *most appropriate* answer from the options given below:

- (1) A and B only
- (2) A, B and C
- (3) B and C only
- (4) B only

165. Read the given below statements about compound epithelium.

- A. This tissue has a limited role in secretion and absorption.
- B. It is mainly involved in formation of diffusion boundary.
- C. It is present in the lining of blood vessels.
- D. It is composed of multilayered cells.





Which of the above statements are **incorrect**?

- (1) A and C only
- (2) B and C only
- (3) C and D only
- (4) A and D only

166. Match List-I with List-II:

| List-I<br>(Placental mammal) | List-II<br>(Australian marsupials) |
|------------------------------|------------------------------------|
|------------------------------|------------------------------------|

- |              |                          |
|--------------|--------------------------|
| (A) Mole     | (I) Spotted cuscus       |
| (B) Lemur    | (II) Tasmanian tiger cat |
| (C) Bobcat   | (III) Marsupial mole     |
| (D) Anteater | (IV) Numbat              |

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-III, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-II, D-IV

167. Read the following statements.

- A. Our laws permit legal adoption.
- B. Legal marriageable age of girls is 16 years.
- C. Legal marriageable age of boys is 18 years.
- D. WHO stands for World Health Organisation.
- E. *Hum do Hamare do* is Indian slogan promoting family planning.

Choose the correct statements from the options given below:

- (1) A, B and D only
- (2) A and B only
- (3) B and C only
- (4) A, D and E

168. Choose the **incorrect** statement w.r.t the sex determination in honeybees.

- (1) Honeybees have a haplodiploid sex-determination system.
- (2) An offspring formed from the union of a sperm and an egg develops as a female.

- (3) The males have half the number of chromosomes than that of a female.
- (4) Male bees have fathers but cannot have sons.

169. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Frogs belong to Class Reptilia.

**Reason R:** Frogs can live both on land and in water.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

170. Select a **mismatch** from the following.

- (1) Stroke volume – 70 ml
- (2) Cardiac output – 5L/min
- (3) End of T-wave – End of diastole
- (4) Coronary artery disease – Atherosclerosis

171. Given below are two statements.

**Statement I:** The symptoms of pneumonia include fever, chills, cough and headache.

**Statement II:** The malarial parasite requires three hosts.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

172. Consider the following characters and identify the disease.
- It is a Mendelian disorder.
  - It is an autosome-linked recessive disease.
  - The defect could be due to either mutation or deletion which ultimately results in reduced rate of synthesis of one of the globin chains ( $\alpha$  and  $\beta$  chains) that make up haemoglobin.
  - The formation of abnormal haemoglobin molecules results in anaemia.

Choose the *most appropriate* answer from the options given below:

- Haemophilia
- Sickle-cell anaemia
- Phenylketonuria
- Thalassemia

173. Match List-I with List-II:

| List-I            | List-II                                |
|-------------------|----------------------------------------|
| (A) MTP           | (I) Relatively safer upto twelve weeks |
| (B) STD           | (II) Oral contraceptive                |
| (C) Vaults        | (III) Genital herpes                   |
| (D) <i>Saheli</i> | (IV) Barrier method                    |

Choose the **correct** answer from the options given below:

- A-III, B-II, C-I, D-IV
- A-I, B-III, C-IV, D-II
- A-II, B-III, C-IV, D-I
- A-III, B-I, C-II, D-IV

174. Removal of proximal convoluted tubule from the nephron will result in:

- no change in quality and quantity of urine.
- no urine formation.
- more diluted urine.
- more concentrated urine.

175. Gene therapy is most successfully used to treat which of the following diseases?

- AIDS
- Cancer
- Cystic fibrosis
- Diabetes

176. Find the **correct** statement about chemical which is obtained from given plant.



- Excessive doses causes hallucinations.
- Abused by some sports person.
- Producing a sense of euphoria.
- These are known for their effects on cardiovascular system of the body.

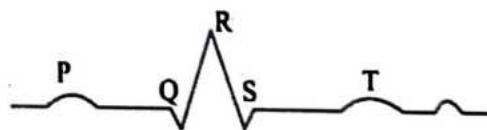
177. Which of the following is an example of a polysaccharide?

- |            |              |
|------------|--------------|
| (1) Ribose | (2) Glucose  |
| (3) Starch | (4) Fructose |

178. Match the columns and choose the correct option.

| Column-I                                   | Column-II     |
|--------------------------------------------|---------------|
| (a) <i>Epidermophyton</i>                  | (i) Bacteria  |
| (b) <i>Wuchereria</i>                      | (ii) Worm     |
| (c) <i>Plasmodium</i>                      | (iii) Fungi   |
| (d) <i>Streptococcus</i>                   | (iv) Protozoa |
| (1) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii) |               |
| (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) |               |
| (3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) |               |
| (4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii) |               |

179. Which wave marks the depolarisation of ventricles?



- P-wave
- Q-wave
- QRS-wave
- T-wave



180. Which of the following statement(s) are correct?

- (a) Ammonia is readily soluble.
  - (b) Loop of Henle is long in cortical nephrons.
  - (c) DCT is lined by simple cuboidal brush border epithelium.
  - (d) Vasa recta is absent or highly reduced in cortical nephrons.
- (1) (a), (b) and (c)  
 (2) (a) and (d)  
 (3) (a), (c) and (d)  
 (4) (b), (c) and (d)

181. Given below are two statements:

**Statement I:** Evolution for Darwin was gradual.

**Statement II:** Fossils are remains of hard parts of life-forms found in rocks.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

182. Which of the following structures are involved in the regulation of sexual behavior and expression of emotional reactions?

- (a) Amygdala
- (b) Cerebral Hemispheres
- (c) Hippocampus
- (d) Hypothalamus
- (e) Medulla oblongata

Choices:

- (1) (a), (b) and (e) only
- (2) (a), (c) and (e) only
- (3) (b), (c) and (d) only
- (4) (a), (b), (c) and (d)

183. The gonadotrophins include:

- (1) progesterone.      (2) FSH
- (3) LH                      (4) both (2) and (3)

184. Match List-I with List-II:

| List-I |                  | List-II |                    |
|--------|------------------|---------|--------------------|
| (A)    | Epidermal scales | (I)     | <i>Torpedo</i>     |
| (B)    | Claspers         | (II)    | <i>Calotes</i>     |
| (C)    | Air bladder      | (III)   | <i>Hippocampus</i> |
| (D)    | Electric organ   | (IV)    | <i>Scoliodon</i>   |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-IV, B-II, C-III, D-I

185. The non-protein organic factor firmly attached to apoenzyme is called:

- (1) metal ion.
- (2) co-enzyme.
- (3) prosthetic group.
- (4) activator.

#### Zoology : Section-B (Q. No. 186 to 200)

186. The chronological order of human evolution from early to the recent is:

- (1) *Ramapithecus* → *Homo habilis*  
→ *Australopithecus* → *Homo erectus*.
- (2) *Australopithecus* → *Homo habilis*  
→ *Ramapithecus* → *Homo erectus*.
- (3) *Australopithecus* → *Ramapithecus* →  
*Homo habilis* → *Homo erectus*.
- (4) *Ramapithecus* → *Australopithecus* →  
*Homo habilis* → *Homo erectus*.

187. Read the following statements:

- A. The reproductive cycle in the female primates is called menstrual cycle.
- B. A functional mammary gland is a characteristic of all female mammals.
- C. The urethra originates from the kidneys and extends through the penis.
- D. Formation of gametes is called gametogenesis.
- E. Myometrium is made up of skeletal muscle.

Choose the *most appropriate* answer from the options given below:

- (1) A, B and E only (2) B and D only  
(3) B, C and E only (4) A, B and D only

188. Refer to the figures given below and identify the structures A, B and C respectively.



A B C

- (1) A: RBCs, B: WBCs, C: Platelets  
(2) A: WBCs, B: Platelets, C: RBCs  
(3) A: Platelets, B: WBCs, C: RBCs  
(4) A: WBCs, B: RBCs, C: Platelets

189. Which structure is responsible for the production of progesterone in the female reproductive system?

- (1) Mammary gland (2) Umbilical cord  
(3) Corpus luteum (4) Uterus

190. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** During gel electrophoresis, the DNA fragments move towards the anode.

**Reason R:** DNA fragments are negatively charged molecules.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.  
(2) A is false but R is true.  
(3) Both A and R are true and R is the correct explanation of A.  
(4) Both A and R are true but R is NOT the correct explanation of A.

191. Match the columns.

**Column-I**

- (a) Gene therapy (i)

**Column-II**

Use of bioresources without proper authorization from the concerned authority

- (b) Probe (ii) Correction of a gene defect  
(c) Biopiracy (iii) A right granted by the government to an inventor to prevent others from using the invention commercially  
(d) Biopatent (iv) ssDNA or RNA with radioactive molecule

- (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)  
(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)  
(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)  
(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

192. Match List-I with List-II:

**List-I**

- (A) Allergy  
(B) T-helper cells  
(C) Hallucinogens  
(D) Liver

**List-II**

- (I) Activation of B-cells  
(II) *Atropa belladonna*  
(III) IgE  
(IV) Cirrhosis

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV  
(2) A-III, B-I, C-II, D-IV  
(3) A-IV, B-II, C-I, D-III  
(4) A-II, B-III, C-IV, D-I

193. Which hormone is primarily responsible for regulating the basal metabolic rate in the human body?

- (1) Insulin (2) Estrogen  
(3) Thyroxine (4) Testosterone

194. Which among the following is a neutral amino acid?

- (1) Lysine  
(2) Valine  
(3) Glutamic acid  
(4) All of these



195. Given below are two statements:

**Statement I:** The most distinctive feature of echinoderms is the presence of water vascular system which helps in locomotion only.

**Statement II:** Reptiles are poikilotherms.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

196. "Frogs exhibit sexual dimorphism". Select the option which justifies the given statement.

- (1) Male frogs have a copulatory pad on the first digit of the forelimbs.
- (2) Presence of sound producing vocal sacs in male frogs.
- (3) Presence of Webbed-digit feet.
- (4) Both (1) and (2)

197. Read the following statements(A-E).

- A. The substrate binds to the active site of the enzyme.
- B. Enzymes isolated from thermophilic organisms get denatured at 50°C.
- C. The active site of enzyme breaks the chemical bonds of the product.

- D. Enzymes generally function in a narrow range of pH.
- E. Catalytic activity is lost when co-factor is removed.

Select the option which includes all correct statements:

- (1) A, B and C only
- (2) B, C and D only
- (3) B, C and E only
- (4) A, D and E only

198. Which of the following structures in a cockroach serves as a grinding mechanism for food?

- (1) Gizzard
- (2) Crop
- (3) Midgut
- (4) Malpighian tubules

199. Match List-I with List-II:

- | List-I            | List-II                      |
|-------------------|------------------------------|
| (A) Amniocentesis | (I) Induced abortion         |
| (B) Saheli        | (II) Detect genetic disorder |
| (C) MTP           | (III) Once a week pill       |
| (D) Nirodh        | (IV) Male condom             |
- Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-II, D-IV

200. The flat bone of axial skeleton lying on the ventral midline of thorax is:

- (1) scapula.
- (2) sternum.
- (3) clavicle.
- (4) diaphragm.